

Network Configuration

Objectives

After completing this lesson, you should be able to:

- Describe network interface configuration files
- Describe additional network configuration files
- Start the network service
- Use the `ethtool` utility
- Describe NetworkManager and use the NetworkManager GUI
- Use the Network Connections editor
- Use the `nmcli` and `nmtui` utilities
- Describe ARP and the ARP cache
- Use the `ip` utility
- Describe networking in Oracle Cloud Infrastructure



Network Interface File Names

- Each physical network device has an associated network interface configuration file.
- Network interface configuration files are located in the `/etc/sysconfig/network-scripts` directory.
- The naming scheme automatically assigns interface names that are predictable, for example:

```
# ls/etc/sysconfig/network-scripts/ifcfg*  
ifcfg-enp134s1f0    ifcfg-enp134s1f1    ifcfg-lo
```

- Names persist across system reboots, hardware reconfiguration, and kernel and device driver updates.
- The naming scheme can be changed by boot parameters:
 - `biosdevname=1`
 - `net.ifnames=0`

Network Interface File Parameters

Configuration parameters include:

- `TYPE=Ethernet`
- `BOOTPROTO=none`
- `DEFROUTE=yes`
- `NAME=eth0`
- `ONBOOT=yes`
- `HWADDR=00:14:4F:8D:B0:BC`
- `IPADDR0=NN.NN.NN.NN`
- `PREFIX0=23`
- `GATEWAY0=NN.NN.NN.NN`
- `DNS1=NN.NN.NN.NN`
- `DNS2=NN.NN.NN.NN`
- `DOMAIN=example.com`



Additional Network Configuration Files

- `/etc/hosts` associates host names with IP addresses.
 - Larger networks would use DNS to perform this resolution.
 - Specify the IP address of the loopback device.
- `/etc/resolv.conf`:
 - Provides access to DNS
 - Identifies DNS name servers and search domain
- `/etc/sysconfig/network` specifies global information for all network interfaces.
- `/etc/nsswitch.conf` lists the order of host name searches.



Starting the Network Service

- Use the `systemctl` command to start, stop, and restart the network service:

```
# systemctl restart network
```

- Interface control scripts in `/etc/sysconfig/network-scripts` can also be used.
- Use `ifup <interface_name>` to activate an interface:

```
# ifup enp134s1f0
```

- Use `ifdown <interface_name>` to deactivate an interface:

```
# ifdown enp134s1f0
```

The `ethtool` Utility

- `ethtool` is used to query and set low-level network interface properties.
- Changes made by `ethtool` do not persist after a reboot.
- To show current low-level properties on an interface:

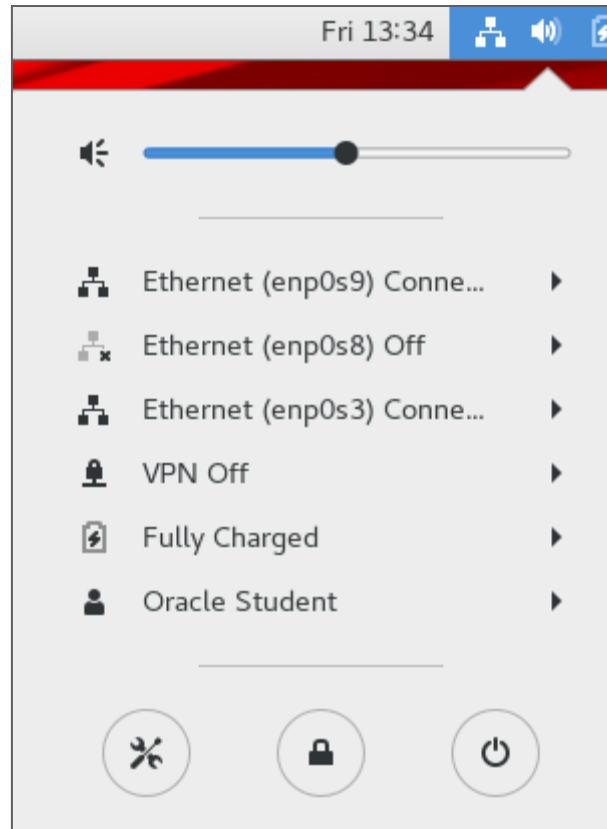
```
# ethtool enp134s1f0  
...
```

- Use the `-s` option to set low-level properties on an interface. Example:

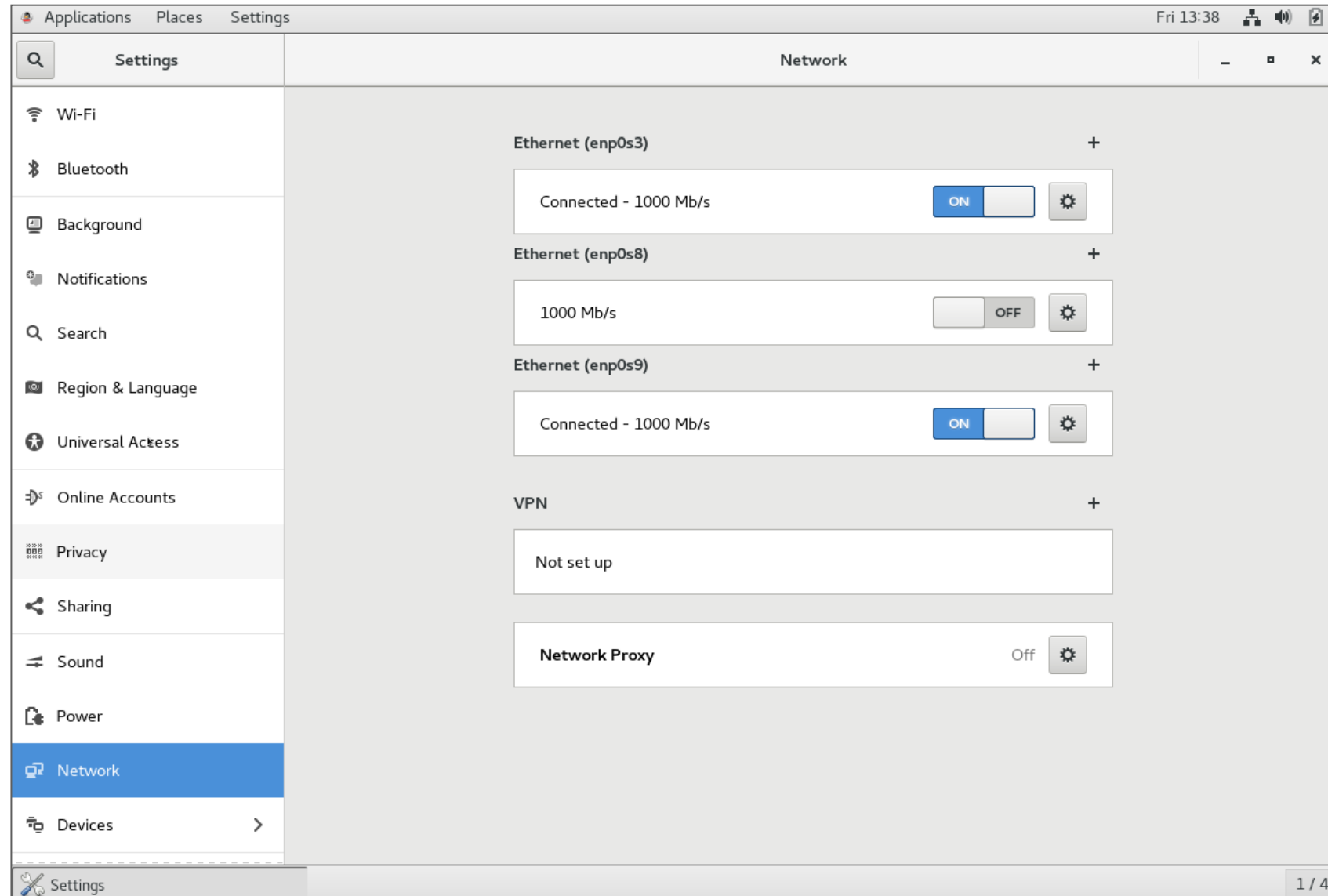
```
# ethtool -s enp134s1f0 speed 1000 autoneg on duplex full
```

NetworkManager

- NetworkManager:
 - Dynamically detects and configures network connections
 - Includes a GNOME Notification Area network icon
- Click the icon to display status and to manage network connections.



Network Settings



View or Edit an Existing Network Connection

Cancel

Wired

Apply

Details

Identity

IPv4

IPv6

Security

Link speed

1000 Mb/s

IPv4 Address

192.168.56.103

IPv6 Address

fe80::9633:5c85:d1db:5200

Hardware Address

08:00:27:CF:08:70

DNS

☒ Connect automatically

☒ Make available to other users

Remove Connection Profile

Network Connections Editor

Network Connections

Name	Last Used
▼ Ethernet	
enp0s9	1 minute ago
enp0s3	1 minute ago
enp0s8	1 minute ago
▼ Bridge	
virbr0	1 minute ago

+ - ⚙

Editing enp0s3

Connection name: enp0s3

General Ethernet 802.1X Security DCB Proxy IPv4 Settings IPv6 Settings

Method: Automatic (DHCP)

Additional static addresses

Address	Netmask	Gateway
---------	---------	---------

Add Delete

Additional DNS servers:

Additional search domains:

DHCP client ID:

☐ Require IPv4 addressing for this connection to complete

Routes...

Cancel Save

The `nmcli` Utility

- This is a command-line tool used to control NetworkManager.
 - Useful for scripting and for controlling NetworkManager without a GUI
- The command provides seven categories, or objects:
 - `general`: NetworkManager's general status and operations
 - `networking`: Overall networking control
 - `radio`: NetworkManager radio switches
 - `connection`: NetworkManager's connections
 - `device`: Devices managed by NetworkManager
 - `agent`: NetworkManager secret and/or polkit agent
 - `monitor`: Monitor NetworkManager changes
- See the `nmcli-examples(5)` manual page for examples.

The nmcli general Object

The nmcli general object provides the following commands:

- status: Show the overall status of NetworkManager.
- hostname: Get or change the system host name. The system host name is stored in /etc/hostname.
- permissions: Show permissions for the various authenticated operations that NetworkManager provides.
- logging: Get or change NetworkManager logging level for domains. See the NetworkManager.conf(5) man page for descriptions of log levels and domains.

The nmcli networking Object

- The nmcli networking object provides the following commands:
 - on: Enable networking by NetworkManager.
 - off: Disable networking by NetworkManager.
 - connectivity [check]: Get network connectivity state.
- To display networking status:

```
# nmcli networking  
enabled
```

- To get network connectivity state:

```
# nmcli networking connectivity check  
full
```

- Possible states: none, portal, limited, full, unknown

The nmcli radio Object

- The nmcli radio object provides the following commands:
 - wifi [on | off]: Get or set status of Wi-Fi in NetworkManager
 - wwan [on | off]: Get or set status of WWAN (mobile broadband)
 - all [on | off]: Get or set status of all the above
- To display status of all the radio switches:

```
# nmcli radio
WIFI-HW  WIFI      WWAN-HW  WWAN
enabled  enabled   enabled   enabled
```

The `nmcli connection` Object

- NetworkManager stores all network configuration information as connections.
- Connections contain all the information required to create or connect to a network.
- There can be multiple connections for a given device.
- Only one connection can be active on a device at a time.
- The `nmcli connection` object provides the following commands, as displayed, by using the `help` argument:

```
# nmcli connection help
Usage: nmcli connection { COMMAND | help }
COMMAND := {show | up | down | add | modify | clone | edit | delete |
            monitor | reload | load | import | export }
```

The nmcli connection show Command

- Use the `show` argument to list connection profiles.
- Include the `--active` option to list only the active profiles.

```
# nmcli connection show --active
```

NAME	UUID	TYPE	DEVICE
eth0	...	ethernet	eth0
virbr0	...	bridge	virbr0

- View detailed information by specifying an `<ID>` keyword and associated value:

```
# nmcli connection show id eth0
```

connection.id:	eth0
...	
connection.type:	802-3-ethernet
connection.autoconnect:	yes
...	

nmcli connection up|down Commands

- Use the `up` argument to activate a connection.
- The connection is specified by its ID, UUID, or D-Bus path.
- The following example uses the connection ID:

```
# nmcli connection up id eth1  
Connection successfully activated (D-Bus active path:  
  /org/freedesktop/NetworkManager/ActiveConnection/13)
```

- Use the `down` argument to deactivate a connection.

```
# nmcli connection down id eth1
```

- If the connection is set to `autoconnect`, the connection starts automatically on the disconnected device again.
- In this case, use the `nmcli device disconnect` command instead of `nmcli connection down`.

The `nmcli connection add` Command

- Use the `add` argument to add a connection for NetworkManager.
- The command accepts the following options:
 - Common options: `type`, `ifname`, `con-name`, and so on
 - Type-specific options: `mac`, `ssid`, `mtu`, and so on
 - IP options: `ip4`, `ip6`, `gw4`, `gw6`
- Example:

```
# nmcli connection add con-name new-eth0 ifname eth0 type ethernet ip4  
192.168.2.100/24 gw4 192.168.2.1  
Connection 'new-eth0' (<UUID>) successfully added.
```

The nmcli connection edit Command

- Use the `edit` argument to edit a connection by using an interactive editor. Example:

```
# nmcli connection edit new-eth0
...
You may edit the following settings: connection, 802-3-ethernet (ethernet),
802-1x, dcb, ipv4, ipv6, proxy
nmcli>
```

- Use the ‘?’ key or type ‘help’ to view commands.
- Use the `edit` argument without specifying a connection to add a new connection. Example:

```
# nmcli connection edit
...
Enter connection type:
...
```

The `nmcli connection modify` Command

- Use the `modify` argument to modify properties in the connection profile.
 - The following example modifies the IPv4 DNS server address property (`ipv4.dns`) for the `new-eth1` connection:

```
# nmcli connection modify new-eth1 ipv4.dns NN.NN.NN.NN
```

- Use the '+' prefix to append a value:

```
# nmcli connection modify new-eth1 +ipv4.dns NN.NN.NN.NN
```

nmcli connection delete | reload | load Commands

- Use the `delete` argument to delete a configured connection. Example:

```
# nmcli connection delete new-eth1
```

- Use the `reload` argument to reload all connection files from disk. Example:

```
# nmcli connection reload
```

- Use the `load` argument to load or reload one or more configuration files from disk. Example:

```
# nmcli connection load /etc/sysconfig/network-scripts/ifcfg-new-eth0
```

- Set `monitor-connection-files` to `true` in a NetworkManager configuration file to cause NetworkManager automatically to reload configuration files when changed.

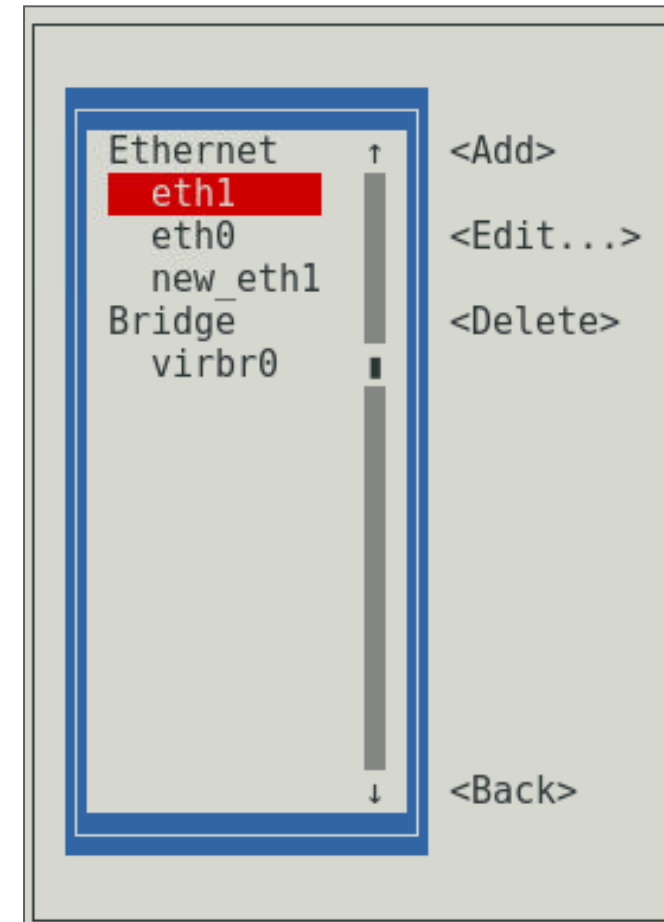
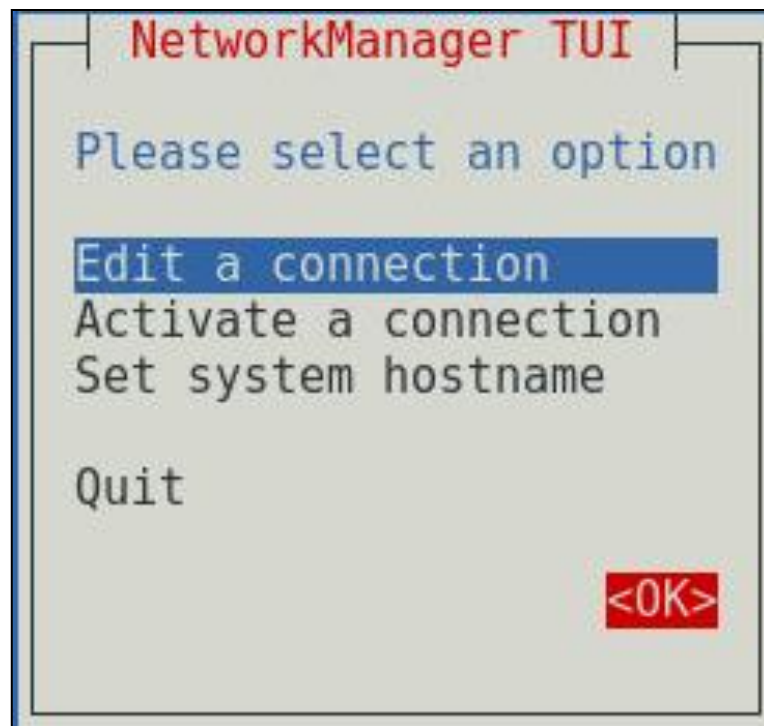
The nmcli device Object

- The `nmcli device` object provides the following commands, among others:
 - `status`: Display the status of all devices.
 - `show [<ifname>]`: Show detailed information about devices.
 - `connect <ifname>`: Connect the device.
 - `disconnect <ifname>`: Disconnect the device.
 - `wifi list | connect | hotspot | rescan`: List Wi-Fi access points, connect to a Wi-Fi network, create a Wi-Fi hotspot, or rescan for available access points.
- To display status of all the devices:

```
# nmcli device
DEVICE          TYPE          STATE          CONNECTION
...
eth0            ethernet     connected      eth0
...
```

The `nmtui` Utility

```
# nmtui
```



The `ip` Utility

- The `ip` utility:
 - Is used to display and manipulate devices, routing, policy routing, and tunnels
 - Replaces the `ifconfig` command
 - Provides several OBJECT arguments, such as:
 - `link`: Network device
 - `address` (or `addr`): IPv4 or IPv6 address on a device
 - `route`: Routing table entry
 - Provides a number of COMMANDS for each OBJECT, such as:
 - `add`, `change`, `del`, `show`, and so on
- Use `help` to show COMMANDS available for an OBJECT.

```
# ip addr help
```

The `ip addr` Object

- Use the `ip addr` object to show and manage IPv4 or IPv6 addresses on a device.
 - Changes made with `ip` are not persistent.

- To show the status of all active devices:

```
# ip addr
```

- To add an IPv4 address to a network interface device:

```
# ip addr add 192.168.2.101/24 dev eth1
```

- To remove an IPv4 address from a device:

```
# ip addr del 192.168.2.101/24 dev eth1
```

- To remove all IP addresses from a device:

```
# ip addr flush dev eth1
```

The `ip link` Object

- Use the `ip link` object to show and manage the attributes of network interfaces on the system.
 - Changes made with `ip` are not persistent.

- To show the status of a specific device:

```
# ip link show eth1
```

- To bring a specific network interface down:

```
# ip link set eth1 down
```

- To bring a specific network interface up:

```
# ip link set eth1 up
```

- To change a device attribute, for example, MTU:

```
# ip link set eth1 mtu 1000
```

Address Resolution Protocol (ARP)

- ARP resolves an IP address to the MAC address.
- IP addresses and associated MAC addresses are cached in an ARP table.
 - By default, entries become stale after 60 seconds.
- Use the `ip neigh` object to display, add, or delete entries in the ARP table.
 - The `arp` command is deprecated.
- To display all entries:

```
# ip neigh  
...
```

- To delete all entries:

```
# ip neigh flush all
```

The `ip route` Object

- Use the `ip route` object to display or manipulate the IP routing table.
- The default route, `GATEWAY`, is configured in the `/etc/sysconfig/network` file.
- To display the routing table:

```
# ip route
```

- To add an entry to the routing table:

```
# ip route add default via NN.NN.NN.2 dev enp134s1f0 proto static  
# ip route add 192.0.2.1 via NN.NN.NN.2 dev enp134s1f0
```

- Configure permanent static routes in the `/etc/sysconfig/network-scripts/route-interface` file.

Networking in Oracle Cloud Infrastructure

- An instance must run within a subnet within a Virtual Cloud Network (VCN).
- VCN contains addresses in the IPv4 CIDR block.
 - Oracle recommends private ranges: `10.0.0.0/8`, `172.16/12`, and `192.168/16`
 - Example CIDR block: `192.168.0.0/16`
 - Publicly routable ranges are possible, but all addresses are termed "private."
 - Comes with a default route table
- Subnet is a subset of VCN, specified as either public or private.
 - CIDR is block specified, for example, `192.168.1.0/24`
 - Private subnet has no public IP addresses.
 - Has a primary virtual network interface card (VNIC), and can have more
 - VNIC allows connection to the VCN.
- Instance is attached to a VNIC within a specified subnet.
 - Public subnet allows SSH connection over the Internet.

Networking in Oracle Cloud Infrastructure

- Virtual route tables allow sending traffic outside a VCN, such as to the Internet or an on-premises network.
 - Default route table has no rules, which can be added later.
 - Route table is only used if destination IP address is outside the VCNs CIDR block.
- Gateways are virtual routers that provide paths for network traffic.
 - For example, an Internet gateway provides access to the Internet
- Secondary VNICs can be added after an instance is launched.
 - Can connect to multiple subnets in a VCN or to multiple VCNs
- Secondary private IP addresses can be added after an instance is launched.
 - Can be used on a standby instance if an instance is not running correctly
- VCN Peering allows direct, private connections between VCNs.
 - Can be local (within a region) or remote (between regions)

Quiz



Which of the following statements is true?

- a. Network interface configuration files are located in the `/etc/sysconfig/network` directory.
- b. The NetworkManager GUI can be used only to configure wired Ethernet devices.
- c. Routing tables can be displayed by using the `netstat -r` command.
- d. The `ipup eth0` command activates the `eth0` interface.



Quiz



Which of the following tasks can you perform by using the `nmcli` utility?

- a. Disable and enable networking operations.
- b. Manage network connection profiles.
- c. Change the system host name.
- d. Manage routing.
- e. Manage the ARP cache.



Quiz



Which of the following tasks can you perform by using the `ip` utility?

- a. Manage network devices.
- b. Manage network addressing.
- c. Change the system host name.
- d. Manage routing.
- e. Manage the ARP cache.



Summary

In this lesson, you should have learned how to:

- Describe network interface configuration files
- Describe additional network configuration files
- Start the network service
- Use the `ethtool` utility
- Describe NetworkManager and use the NetworkManager GUI
- Use the Network Connections editor
- Use the `nmcli` and `nmtui` utilities
- Describe ARP and the ARP cache
- Use the `ip` utility
- Describe networking in Oracle Cloud Infrastructure



Practice 13: Overview

This practice covers the following topics:

- Configuring the `eth1` network interface
- Using NetworkManager with the GNOME GUI
- Using the Network Connection Editor
- Using the `nmcli` utility
- Using the `nmtui` utility
- Using the `ip` utility